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CIRCUIT BOARD DEVICE

Abstract

A circuit board device for transmitting signals at a frequency of 50 GHz to 300 GHz is provided. The circuit board device includes an insulating portion, first and second interlayer conduction structures passing through the insulating portion, a first reference layer, and a second reference layer. The first interlayer conduction structure defines a signal path and includes a first pillar including first end portions and a first central section. The first thickness of the first end portions is greater than the second thickness of the first central section. The first and second reference layers are located on top and bottom surfaces of the insulating portion and surround the first interlayer conduction structure. Two opposite second end surfaces of the second interlayer conduction structure are respectively connected to the first reference layer and the second reference layer to define a ground path surrounding the signal path.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
[0001] This application claims priority of Taiwan Patent Application No. 113108801, filed on Mar. 11, 2024, the entirety of which is incorporated by reference herein.
BACKGROUND OF THE INVENTION
Field of the Invention
[0002] The present invention relates to a circuit board device, in particular to an interlayer connective structure of a circuit board device.
Description of the Related Art
[0003] With the rapid development of electronic devices, circuit boards in electronic devices need to quickly transmit current signals at higher frequencies. However, when current flows through the interlayer connective structure in the circuit board, the current signal may be affected by characteristic differences, such as the structure or interfaces of the interlayer connective structure. Several problems can occur, such as impedance mismatch, high reflection loss and high insertion loss, thereby affecting the integrity of high-frequency signals. In order to comply with the current development trend of electronic devices, the signal transmission quality of the conventional circuit boards must be improved.
BRIEF SUMMARY OF THE INVENTION
[0004] An embodiment of the present invention provides a circuit board device for transmitting signals at a frequency of 50 GHz to 300 GHz. The circuit board device includes an insulation portion, a first interlayer connective structure, a first reference layer, a second reference layer, and at least one second interlayer connective structure. The insulation portion has a top surface and a bottom surface. The first interlayer connective structure passes through the insulation portion and defines a signal path. The first interlayer connective structure includes a first pillar. The first pillar passes through the insulation portion and has two opposite first end surfaces. The first pillar includes a pair of first end portions and a first central section. The pair of first end portions include the respective first end surfaces. The first central section is connected between the first end portions. The first wall thickness of each of the first end portions is greater than a second wall thickness of the first central section. The first reference layer is located on the top surface of the insulation portion. The first reference layer is separated from the first interlayer connective structure and surrounds the first interlayer connective structure. The second reference layer is located on the bottom surface of the insulation portion. The second reference layer is separated from the first interlayer connective structure, and surrounds the first interlayer connective structure. The second interlayer connective structure passes through the insulation portion. Two opposite second end surfaces of the second interlayer connective structure are respectively connected to the first reference layer and the second reference layer to define a ground path that surrounds the signal path. An upper surface and a lower surface of the circuit board device include the respective first end surfaces.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The present invention can be more fully understood by reading the subsequent detailed description and examples with references made to the accompanying drawings, wherein:

[0006] FIG. **1**A is a schematic perspective view of a circuit board device in accordance with one embodiment of the disclosure, showing the arrangement of an interlayer connective structure of the circuit board device;

[0007] FIG. **1**B is a schematic top view of the circuit board device of FIG. **1**A, showing a schematic top view of an interlayer connective structure of a circuit board device in accordance with one embodiment of the disclosure;

[0008] FIG. **1**C is a schematic cross-sectional view along line A-A′ of the circuit board device of FIG. **1**A, showing a schematic cross-sectional view of an interlayer connective structure of a circuit board device in accordance with one embodiment of the disclosure;

[0009] FIG. **2** is a schematic top view of an interlayer connective structure of a circuit board device in accordance with one embodiment of the disclosure;

[0010] FIG. **3**A is a schematic perspective view of a circuit board device in accordance with one embodiment of the disclosure, showing the arrangement of an interlayer connective structure of the circuit board device;

[0011] FIG. **3**B is a schematic top view of FIG. **3**A, showing a schematic top view of an interlayer connective structure of a circuit board device in accordance with one embodiment of the disclosure;

[0012] FIG. **3**C is a schematic cross-sectional view along line A-A′ of the circuit board device of FIG. **3**A, showing a schematic cross-sectional view of an interlayer connective structure of a circuit board device in accordance with one embodiment of the disclosure;

[0013] FIG. **4** is a schematic top view showing an interlayer connective structure of a printed circuit board of a comparative example;

[0014] FIG. **5** is a schematic top view showing an interlayer connective structure of a printed circuit board of a comparative example;

[0015] FIG. **6** is a schematic top view showing an interlayer connective structure of a printed circuit board of a comparative example;

[0016] FIG. **7** is a diagram showing the comparison of the simulations of terminal time-domain reflectometry (TDR) characteristic impedance versus signal transmission time between an interlayer connective structure of a circuit board device in accordance with one embodiment of the disclosure and an interlayer connective structure of a printed circuit board of a comparative example; and

[0017] FIG. **8** is a diagram showing the comparison of the simulations of reflection coefficient (return loss, S.sub.11 (dB)) versus signal frequency (GHz) between an interlayer connective structure of a circuit board device in accordance with one embodiment of the disclosure and an interlayer connective structure of a printed circuit board of the comparative example; and

[0018] FIG. **9** is a diagram showing the comparison of the simulations of insertion loss (S.sub.21 (dB)) versus signal frequency (GHz) between an interlayer connective structure of a circuit board device in accordance with one embodiment of the disclosure and an interlayer connective structure of a printed circuit board of the comparative example.

DETAILED DESCRIPTION OF THE INVENTION

[0019] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0020] The spatially relative terms mentioned herein, such as “upper”, “lower”, “left”, “right” and the like refer to the orientation depicted in the figures. Accordingly, the spatially relative terms are intended to illustrate and are not intended to be limiting.

[0021] In some embodiments of the disclosure, the terms such as “dispose”, “connect” and the like refer to arrangement and connection, unless otherwise specified, may refer to embodiments in which the two features are in direct contact, and may also refer to embodiments in which additional features may be located between the two features, such that the twos features may not be in direct contact. The terms refer to arrangement and connection may also include the embodiments in which both structures are movable, or both structures are fixed.

[0022] In addition, the terms “first”, “second” and the like mentioned in this specification or the claims are used to name different features or to distinguish different embodiments or ranges, and are not used to limit the upper limit or lower limit of the number of features, and are also not intended to limit the order of manufacture or arrangement of feature.

[0023] For the purpose of fully understanding the features and advantages of the disclosure, the subsequent specific embodiments of the disclosure are described in detail with references made to the accompanying drawings.

[0024] FIG. **1**A is a schematic perspective view of a circuit board device **500**A in accordance with one embodiment of the disclosure, showing the arrangement of a first interlayer connective structure **210** and a second interlayer connective structure **230** of the circuit board device **500**A. FIG. **1**B is a schematic top view of the circuit board device **500**A of FIG. **1**A, showing a schematic top view of the first interlayer connective structure **210** and the second interlayer connective structure **230** of the circuit board device **500**A in accordance with one embodiment of the disclosure. FIG. **1**C is a schematic cross-sectional view along line A-A′ of the circuit board device **500**A of FIG. **1**A, showing a schematic cross-sectional view of the first interlayer connective structure **210** and the second interlayer connective structure **230** of the circuit board device **500**A in accordance with one embodiment of the disclosure. In some embodiments, the circuit board device **500**A includes a printed circuit board for transmitting signals at a frequency of 50 GHz to 300 GHz. Moreover, the first interlayer connective structure **210** for transmitting signals has a pad-free design.

[0025] As shown in FIGS. **1**A, **1**B, and **1**C, for illustration, the circuit board device **500**A is a double-sided printed circuit board as an example, in which from the topmost layer to the bottommost layer are respectively marked with a level L1 and a level L2 in sequence. Furthermore, each level may include at least one wiring layer. However, the disclosure may also be applied to other printed circuit board devices with different numbers of layers. Moreover, directions D**1** and D**2** labeled in FIGS. **1**A, **1**B and **1**C are defined as the horizontal directions (the directions D**1** and D**2** also serve as the extending directions of the conductive lines/transmission lines), and a direction D**3** is defined as the vertical direction (or the extending direction of the plating through hole).

[0026] In some embodiments, the circuit board device **500**A includes a substrate **200**, the first interlayer connective structure **210**, a reference layer **220**T, a reference layer **220**B, and the second interlayer connective structure **230**. In this embodiment, the first interlayer connective structure **210** of the circuit board device **500**A is a plating through hole (PTH) structure having a pad-free design. That is to say, the first interlayer connective structure **210** used for transmitting signals has opposite end surfaces exposed from the substrate **200**. The opposite end surfaces are not directly covered by or directly connected to pads. In addition, the through via **214** of the first interlayer connective structure **210** is not sealed by pads. For illustration, the substrate **200** in FIGS. **1**A and **1**B is shown with a dotted line.

[0027] In some embodiments as shown in FIGS. **1**A, **1**B, and **1**C, the substrate **200** may include an insulation portion **202** and conductive layers (not shown) located at the level L1 and the level L2. For example, the conductive layer located at the level L1 includes a reference layer **220**T and a wiring layer (not shown) located on the top surface **202**T of the insulation portion **202**. The conductive layer located at level L2 includes a reference layer **220**B and another wiring layer (not shown) located on the bottom surface **202**B of the insulation portion **202**. Moreover, the insulation portion **202** is located between the reference layer **220**T, the reference layer **220**B and the two wiring layers.

[0028] In other embodiments, the substrate **200** may include a multilayer printed circuit board and have more than two wiring layers, in which at least one wiring layer may be located within the insulation portion **202**. More specifically, the insulation portion **202** may include multiple insulating layers stacked on each other (not shown), and at least one wiring layer may be sandwiched between adjacent two insulating layers.

[0029] In some embodiments, the insulation portion **202** may be or may include prepreg (PP) containing polymer materials, fiber materials, or other suitable materials. However, the disclosure is not limited to the disclosed embodiments. For example, the polymer material may be or include epoxy resin, polyimide (PI), other suitable polymer materials or a combination thereof. However, the disclosure is not limited to the disclosed embodiments. The fiber material may include carbon fiber, glass fiber, other suitable fiber materials or a combination thereof. However, the disclosure is not limited to the disclosed embodiments. In this embodiment, the insulation portion **202** may be formed of prepreg (PP), such as EM-890K material from Elite Material Co., Ltd. In some embodiments, a thickness T**1** of the insulation portion **202** may be in a range of 2.09 mm to 2.29 mm, such as about 2.19 mm.

[0030] The first interlayer connective structure **210** is formed passing through the insulation portion **202** and defines a signal path of the circuit board device **500**A. In other words, the first interlayer connective structure **210** is used to transmit signals. In some embodiments, the first interlayer connective structure **210** includes a first pillar **212**. More specifically, the first interlayer connective structure **210** is located in the through hole **204** passing through the insulation portion **202**. Moreover, the first pillar **212** covers and is in contact with an inner wall **204**S of the through hole **204**. The through hole **204** has a diameter R**1** (which may also serve as the outer diameter of the first pillar **212**). In some embodiments, the diameter R**1** is in a range of 100 μm to 200 μm, such as about 150 μm.

[0031] As shown in FIG. **1**C, the first pillar **212** has two opposite first end surfaces **212**E**1** and **212**E**2**. Furthermore, the first end surfaces **212**E**1** and **212**E**2** are respectively close to the top surface **202**T and the bottom surface **202**B of the insulation portion **202**. In this embodiment, the first interlayer connective structure **210** may be hollow. More specifically, the first pillar **212** may have a tubular shape and have a through via **214**. The through via **214** and the through hole **204** may be substantially coaxial. In addition, the axes of the through via **214** and the through hole **204** may overlap the first pillar axis A**212** of the first pillar **212**. In some embodiments, the first pillar **212** includes a pair of first end portions **212**T**1**, **212**T**2** and a first central section **212**C. The first end portions **212**T**1** and **212**T**2** are respectively close to the top surface **202**T and the bottom surface **202**B of the insulation portion **202**. Furthermore, the first central section **212**C is connected between the first end portions **212**T**1** and **212**T**2**. The first end portions **212**T**1** and **212**T**2** respectively have the first end surfaces **212**E**1** and **212**E**2**.

[0032] As shown in FIG. **1**B, the tubular first pillar **212** has a first pillar inner diameter R**2**. In some embodiments, the first pillar inner diameter R**2** is in a range of 50 μm to 150 μm, for example, about 100 μm. The outer diameter of the first pillar **212** may be the same as the diameter R**1** of the through hole **204**. In some embodiments, the outer diameter of the first pillar **212** is in a range of 100 μm to 200 μm, for example, about 150 μm.

[0033] It is noted that although, the first interlayer connective structure **210** may be hollow in this embodiment, the first interlayer connective structure **210** may also be solid in other embodiments. More specifically, the first pillar **212** in other embodiments may be a solid pillar without the through via **214**. Therefore, the first interlayer connective structure **210** is not limited to be hollow. In addition, in other embodiments, the first interlayer connective structure **210** may also include a filling material that fills up the through via **214**, such as ink, silver glue, or copper paste.
[0034] In some embodiments, wall thicknesses T**11** and T**12** of the first end portions **212**T**1** and **212**T**2** may be greater than a wall thickness TIC of the first central section **212**C. The wall thickness TIC shown in FIG. **1**C may be the thinnest wall thickness of the first pillar **212**. In other words, the thinnest wall thickness (i.e., the wall thickness TIC shown in FIG. **1**C) of the first pillar **212** is located at the first central section **212**C. In addition, the wall thicknesses T**11** and T**12** of the first end portions **212**T**1** and **212**T**2** may decrease gradually from the corresponding first end surface **212**E**1** and **212**E**2** to the first central section **212**C, as shown in FIG. **1**C. Therefore, the through via **214** has a non-uniform hole diameter (i.e., the first pillar inner diameter R**2**). In some embodiments, the minimum value of the wall thicknesses T**11**, T**12** and the wall thickness TIC is, for example, about 25 μm.
[0035] In some embodiments, the first interlayer connective structure **210** has a pad-free design, which means that the first pillar **212** has two opposite first end surfaces **212**E**1** and **212**E**2** that are not directly covered by or directly connected to the pads. In other words, the opposite upper surface **500**TS and the lower surface **500**BS of the circuit board device **500**A include opposing first end surfaces **212**E**1** and **212**E**2** respectively. Also, the through via **214** is not sealed by pads.
[0036] The reference layers **220**T and **220**B (also called as ground layers **220**T and **220**B) are respectively located on the top surface **202**T and the bottom surface **202**B of the insulation portion **202**. The reference layers **220**T and **220**B are respectively separated from the first interlayer connective structures **210**. In addition, the reference layers **220**T and **220**B surround a first number of first interlayer connective structures **210** (and the first number is a positive integer). In some embodiments, the first number is equal to 1.
[0037] In some embodiments, the reference layers **220**T and **220**B have the same size and top view shape. For example, the reference layers **220**T and **220**B may have an annular shape or other similar shapes. The projection (the vertical projection) of the reference layer **220**T in the direction D**3** (FIGS. **1**A and **1**C) may completely overlap the reference layer **220**B. In other words, in the top view shown in FIG. **1**C, the reference layer **220**T completely overlaps the reference layer **220**B. Moreover, a center **220**TC (a center of mass) of mass of the reference layer **220**T and a center (a center of mass) **220**BC of the reference layer **220**B may both overlap the first pillar axis A**212** of the first pillar **212** in the top view shown in FIG. **1**C.
[0038] As shown in FIG. **1**A, the reference layers **220**T and **220**B have reference layer inner diameters R**3** and R**4**, respectively. In some embodiments, the reference layer inner diameters R**3** and R**4** have the same value as each other. In addition, the reference layer inner diameters R**3** and R**4** are in a range of 500 μm to 600 μm, for example, about 550 μm.
[0039] As shown in FIGS. **1**A and **1**B, the reference layers **220**T and **220**B have reference layer outer diameters R**5** and R**6**, respectively. In some embodiments, the reference layer outer diameters R**5** and R**6** have the same value as each other. In addition, the reference layer outer diameters R**5** and R**6** are in a range of 1200 μm to 1300 μm, for example, about 1250 μm. In some embodiments, a radial width W**1** (W**1**−0.5*(R**5**−R**3**) or W**1**=0.5*(R**6**−R**4**)) of the reference layers **220**T and **220**B is in a range of 300 μm to 500 μm, for example 350 μm.
[0040] As shown in FIGS. **1**A, **1**B, and **1**C, in this embodiment, a plurality of second interlayer connective structures **230** is formed passing through the insulation portion **202**. Moreover, each of the second interlayer connective structures **230** has two opposite second end surfaces **232**E**1** and **232**E**2**. The second end surfaces **232**E**1** and **232**E**2** are respectively connected to the reference layers **220**T and **220**B to define a ground path of the circuit board device **500**A. The ground path

may surround the signal path defined by the first interlayer connective structure **210**. Therefore, the second interlayer connective structure **230** may also be called as a ground plating through hole structure **230**. Furthermore, the reference layers **220**T and **220**B may be both connected to a second number of second interlayer connective structures **230** (and the second number is a positive integer). In some embodiments, the second number is in a range of greater than or equal to 4. In this embodiment, the second number is equal to 4.

[0041] In some embodiments, the second interlayer connective structure **230** includes a second pillar **232**. Moreover, the second pillar **232** covers and is in contact with an inner wall **206**S of the corresponding through hole **206**. As shown in FIG. **1**B, the through hole **206** has a diameter R**7** (which may also serve as the outer diameter of the second pillar **232**), which has the same value as the diameter R**1** of the through hole **204**. In some embodiments, the diameter R**7** is in a range of 100 μm to 200 μm, such as about 150 μm.

[0042] As shown in FIG. **1**B, the through hole **204** is spaced apart from the through hole **206** by a first distance LD**1**. In some embodiments, the first distance LD**1** is in a range of 250 μm to 350 μm, for example, about 300 μm.

[0043] In the top view shown in FIG. **1**B, the through hole **206** is spaced apart from an inner edge **220**TE**1** of the reference layer **220**T (or an inner edge of the reference layer **220**B) by a second distance LD**2**. In some embodiments, the second distance LD**2** is in a range of 50 μm to 150 μm, such as about 100 μm. Moreover, as shown in FIG. **1**B, the through hole **206** is spaced apart from an outer edge **220**TE**2** of the reference layer **220**T (or an outer edge of the reference layer **220**B) by a third distance LD**3**. In some embodiments, the second distance LD**2** and the third distance LD**3** have the same value. For example, the third distance LD**3** may be in a range of 50 μm to 150 μm, such as about 100 μm.

[0044] In some embodiments, through-hole axes A**206** of the through holes **206** are located on a circle whose center is located at a first through-hole axis A**204** of the through hole **204**. In some embodiments, a central angle A**1** between two radii connecting the through-hole axes A**206** of the two closest through holes **206** to the through-hole axis A**204** of the through hole **204** is less than or equal to 90 degrees. For example, as shown in FIG. **1**B, the circuit board device **500**A has four through holes **206**, and the central angle A**1** between two radii connecting the through-hole axes A**206** of the two closest through holes **206** to the through-hole axis A**204** of the through hole **204** is equal to 90 degrees.

[0045] In some embodiments, the first pillar **212** of the first interlayer connective structure **210** and the second pillar **232** of the second interlayer connective structure **230** have the same or similar shape, structure, and size. The second end surfaces **232**E**1** and **232**E**2** of the second pillar **232** of the second interlayer connective structure **230** are respectively close to the top surface **202**T and the bottom surface **202**B of the insulation portion **202**. In this embodiment, the second interlayer connective structure **230** may be hollow. More specifically, the second pillar **232** may have a tubular shape and have a through via **216**. The through via **216** and the through hole **206** may be substantially coaxial. In addition, the axes of the through via **216** and the through hole **206** may overlap the second pillar axis A**232** of the second pillar **232**. In some embodiments, the second pillar **232** includes a pair of second end portions **232**T**1**, **232**T**2** and a second central section **232**C. The second end portions **232**T**1** and **232**T**2** are respectively close to the top surface **202**T and the bottom surface **202**B of the insulation portion **202**. Furthermore, the second central section **232**C is connected between the second end portions **232**T**1** and **232**T**2**. The second end portions **232**T**1** and **232**T**2** respectively have the second end surfaces **232**E**1** and **232**E**2**.

[0046] As shown in FIG. **1**B, the tubular second pillar **232** has a second pillar inner diameter R**8**, which has the same value as the first pillar inner diameter R**2** of the first pillar **212**. In some embodiments, the second pillar inner diameter R**8** is in a range of 50 μm to 150 μm, for example, about 100 μm. The outer diameter of the second pillar **232** may be the same as the diameter R**7** of the through hole **206** and may be in a range of 100 μm to 200 μm, for example, about 150 μm.

[0047] In some embodiments, wall thicknesses T**31** and T**32** of the second end portions **232**T**1** and **232**T**2** may be greater than a wall thickness T**3**C of the second central section **232**C. The wall thickness T**3**C shown in FIG. **1**C may be the thinnest wall thickness of the second pillar **232**. In other words, the thinnest wall thickness (i.e., the wall thickness T**3**C shown in FIG. **1**C) of the second pillar **232** is located at the second central section **232**C. In addition, the wall thicknesses T**31**, T**32** of the second end portions **232**T**1**, **232**T**2** may decrease gradually from the corresponding second end surface **232**E**1** and **232**E**2** to the second central section **232**C, as shown in FIG. **1**C. Therefore, the through via **216** has a non-uniform hole diameter. In some embodiments, the wall thicknesses T**31**, T**32** and the wall thicknesses T**11**, T**12** may have the same thickness. In addition, the wall thickness T**3**C and the wall thickness TIC may have the same thickness. In some embodiments, the minimum value of the wall thicknesses T**31**, T**32** and the wall thickness T**3**C is, for example, about 25 μm.
[0048] It is noted that although the second interlayer connective structure **230** may be hollow in this embodiment, the second interlayer connective structure **230** may also be solid in other embodiments. More specifically, the second pillar **232** in other embodiments may be a solid pillar without the through via **216**. Therefore, the second interlayer connective structure **230** is not limited to be hollow. In addition, in other embodiments, the second interlayer connective structure **230** may also include a filling material that fills up the through via **216**, such as ink, silver glue, or copper paste.
[0049] In some embodiments, the first interlayer connective structure **210** may further include a seed layer **211** located within the through hole **204**. The reference layers **220**T and **220**B may further include a seed layer **219** between them and the top surface **202**T and the bottom surface **202**B of the insulation portion **202**. The second interlayer connective structure **230** may further include a seed layer **231** located within the through hole **206**.
[0050] FIG. **2** is a schematic top view showing the first interlayer connective structure **210** and the second interlayer connective structure **230** of the circuit board device **500**B in accordance with one embodiment of the disclosure, and the reference numbers the same or similar as those previously described with reference to FIG. **1** denote the same or similar elements. At least one difference between the circuit board device **500**A (FIGS. **1**A, **1**B and **1**C) and the circuit board device **500**B (FIG. **2**) is that the reference layer **220**T and the reference layer **220**B of the circuit board device **500**B are both connected to 6 second conduction structures **230**. As shown in FIG. **2**, the central angle A**2** between two radii connecting the through-hole axes A**206** of the two closest through holes **206** to the through-hole axis A**204** of the through hole **204** may be less than or equal to 90 degrees, for example, about 60 degrees.
[0051] FIG. **3**A is a schematic perspective view of a circuit board device **500**C in accordance with one embodiment of the disclosure, showing the arrangement of the first interlayer connective structure **210** and a second interlayer connective structure **330** of the circuit board device **500**C. FIG. **3**B is a schematic top view of FIG. **3**A, showing a schematic top view of the first interlayer connective structure **210** and the second interlayer connective structure **330** of the circuit board device **500**C in accordance with one embodiment of the disclosure. FIG. **3**C is a schematic cross-sectional view along line A-A′ of the circuit board device **500**C of FIG. **3**A, showing the first interlayer connective structure **210** and the second interlayer connective structure **330** of the circuit board device **500**C in accordance with one embodiment of the disclosure. In FIGS. **3**A, **3**B, and **3**C, the reference numbers the same or similar as those previously described with reference to FIGS. **1**A, **1**B, **1**C, and **2** denote the same or similar elements. At least one difference between the circuit board device **500**A (FIGS. **1**A, **1**B and **1**C) (or the circuit board device **500**B (FIG. **2**)) and the circuit board device **500**C is that the circuit board device **500**C includes a coaxial via structure **350** formed by the first interlayer connective structure **210** and the second interlayer connective structure **330**. Furthermore, the first interlayer connective structure **210** may serve as the inner conductor structure of the coaxial via structure **350**. In addition, the second interlayer connective

structure **330** surrounding the first interlayer connective structure **210** may serve as the outer conductor structure of the coaxial via structure **350**. In some embodiments, the impedance of the coaxial perforated structure **350** is in a range of 45 ohms (52) to 55 ohms, such as about 50.7 ohms. In addition to the first interlayer connective structure **210** and the second interlayer connective structure **330**, the coaxial via structure **350** further includes a dielectric material **338** (such as resin, ink, silver glue or copper paste) located between the first interlayer connective structure **210** and the second interlayer connective structure **330**.

[0052] As shown in FIGS. **3**A, **3**B, and **3**C, the second interlayer connective structure **330** (also serve as the ground plating through hole structure **330**) is located in the through hole **306** passing through the insulation portion **202**. In some embodiments, the second interlayer connective structure **330** includes a second pillar **332**. In some embodiments, the second pillar **232** of the second interlayer connective structure **230** (FIGS. **1**A, **1**B, and **1**C) and the second pillar **332** of the second interlayer connective structure **330** have the same or similar structure.

[0053] More specifically, the second interlayer connective structure **330** is located in the through hole **306** passing through the insulation portion **202**. Moreover, the second pillar **332** covers and is in contact with the inner wall **306**S of the corresponding through hole **306**. As shown in FIG. **3**B, the through hole **306** has a diameter R**9** (which may also serve as the outer diameter of the second pillar **332**). In some embodiments, the diameter R**9** is in a range of 650 μm to 750 μm, such as about 700 μm.

[0054] Similar to the second interlayer connective structure **230**, the second pillar **332** of the second interlayer connective structure **330** may have two opposite second end surfaces **332**E**1** and **332**E**2**. Furthermore, the second end surfaces **332**E**1** and **332**E**2** are respectively close to the top surface **202**T and the bottom surface **202**B of the insulation portion **202**. In this embodiment, the second pillar **332** may have a tubular shape and have a through via **316**. The through via **316** and the through hole **306** may be substantially coaxial. In addition, the axes of the through via **316** and the through hole **306** may overlap a second pillar axis A**332** of the second pillar **332**. In some embodiments, the second pillar **332** includes a pair of second end portions **332**T**1**, **332**T**2** and a second central section **332**C. The second end portions **332**T**1** and **332**T**2** are respectively close to the top surface **202**T and the bottom surface **202**B of the insulation portion **202**. Furthermore, the second central section **332**C is connected between the second end portions **332**T**1** and **332**T**2**. The second end portions **332**T**1** and **332**T**2** respectively have the second end surfaces **332**E**1** and **332**E**2**.

[0055] As shown in FIG. **3**B, the tubular second pillar **332** has a second pillar inner diameter R**10**. In some embodiments, the second pillar inner diameter R**10** is in a range of 600 μm to 700 μm, for example, about 650 μm. The outer diameter of the second pillar **332** may be the same as the diameter R**9** of the through hole **306**. In addition, the outer diameter of the second pillar **332** in a range of 650 μm to 750 μm, for example, about 700 μm.

[0056] In some embodiments, wall thicknesses T**71** and T**72** of the second end portions **332**T**1** and **332**T**2** may be greater than a wall thickness T**7**C of the second central section **232**C. The wall thickness T**7**C shown in FIG. **3**C may be the thinnest wall thickness of the second pillar **332**. In other words, the thinnest wall thickness (i.e., the wall thickness T**7**C shown in FIG. **3**C) of the second pillar **332** is located at the second central section **332**C. In addition, the wall thicknesses T**71** and T**72** of each of the second end portions **332**T**1** and **332**T**2** may decrease gradually from the corresponding second end surface **332**E**1** and, **332**E**2** to the second central section **332**C, as shown in FIG. **3**C. Therefore, the through via **316** has a non-uniform hole diameter. In some embodiments, the wall thicknesses T**71**, T**72** and the wall thicknesses T**11**, T**12**, T**31**, T**32** may have the same thickness. In addition, the wall thickness T**7**C and the wall thicknesses TIC, T**3**C may have the same thickness. In some embodiments, the minimum value of the wall thicknesses T**71**, T**72** and the wall thickness T**7**C is, for example, about 25 μm.

[0057] As shown in FIGS. **3**A, **3**B, and **3**C, the dielectric material **338** of the circuit board device

500C is located in the second interlayer connective structure **330** and fills the through hole **306** and the through via **316**. Furthermore, the second pillar **332** surrounds the dielectric material **338**. The first interlayer connective structure **210** is located in the through hole **304** passing through the dielectric material **338**. Moreover, the first pillar **212** covers an inner wall **304**S of the through hole **304**. The through hole **304** and the through hole **306** may be substantially coaxial. In addition, the axes of the through hole **304** and the through hole **306** may overlap the first pillar axis A**212** of the first pillar **212** and the second pillar axis A**332** of the second pillar **332**. The through hole **204** (FIGS. **1**B and **1**C) and the through hole **304** may have the same diameter R**1** (which may also serve as the outer diameter of the first pillar **212**). In some embodiments, the diameter of the through hole **304** is in a range of 100 μm to 200 μm, such as about 150 μm. In some embodiments, the through hole **304** is spaced apart from the through hole **306** by a fourth distance LD**4**. In addition, the fourth distance LD**4** is in a range of 225 μm to 325 μm, for example, about 275 μm.
[0058] The circuit board arrangement **500**C also includes reference layers **320**T and **320**B. The reference layers **320**T and **320**B are respectively located on the top surface **202**T and the bottom surface **202**B of the insulation portion **202**. The reference layers **320**T and **320**B respectively partially cover the top surface **338**T and the bottom surface **338**B of the dielectric material **338** and extend toward the first interlayer connective structure **210**. The reference layers **320**T and **320**B are respectively spaced apart from the first interlayer connective structures **210** and surround a first number of first interlayer connective structures **210**. Furthermore, the reference layers **320**T and **320**B are respectively connected to a second number of second interlayer connective structures **330** (the first number and the second number are positive integers). In this embodiment, the first number and the second number are both equal to 1.
[0059] Similar to reference layers **220**T, **220**B (FIGS. **1**A, **1**B, **1**C), the reference layers **320**T, **320**B may have the same size and top view shape as each other. Moreover, the centers (centers of mass) (not shown) of the reference layers **320**T and **320**B both overlap the first pillar axis A**212** of the first pillar **212**.
[0060] As shown in FIGS. **3**A and **3**B, the reference layers **320**T and **320**B have reference layer inner diameters R**11** and R**12** respectively. In some embodiments, the reference layer inner diameters R**11** and R**12** have the same numerical value as each other. In addition, the reference layer inner diameters R**11** and R**12** are in a range of 475 μm to 525 μm, for example, about 500 μm.
[0061] As shown in FIGS. **1**A and **1**B, the reference layer outer diameters of the reference layers **320**T and **320**B may be the same as the diameter R**9** of the through hole **306**. In some embodiments, the reference layer outer diameter of the reference layers **320**T, **320**B is in a range of 650 μm to 750 μm, for example, about 700 μm. In some embodiments, a radial width W**2** (W**2**=0.5*(R**9**−R**11**) or W**2**−0.5*(R**9**−R**12**)) of the reference layers **320**T, **320**B is in a range of 75 μm to 125 μm, for example, about 100 μm.
[0062] In some embodiments, the first interlayer connective structure **210** may further include a seed layer **211** located within the through hole **304**. The reference layers **320**T, **320**B may further include a seed layer **319** between them and the top surface **202**T and the bottom surface **202**B of the insulation portion **202**. The second interlayer connective structure **230** may further include a seed layer **331** located within the through hole **206**.
[0063] The comparison of the simulations of terminal time-domain reflectometry (TDR) characteristic impedance versus signal transmission time, reflection coefficient (return loss, S.sub.11 (dB)) versus signal frequency (GHz) and insertion loss (S.sub.21 (dB)) versus signal frequency (GHz) between the interlayer connective structures of the circuit board devices **500**A, **500**B and **500**C in accordance with some embodiments of the disclosure and interlayer connective structures of printed circuit boards **700**A, **700**B, **700**C of comparative examples shown in FIGS. **4**, **5** and **6** will be described with reference to FIGS. **7**, **8** and **9**. The reference numbers of the printed circuit boards **700**A, **700**B, **700**C of the comparative examples shown in FIGS. **4**, **5** and **6** that are the same or similar to those in FIGS. **1**A, **1**B, **1**C, **2**, **3**A, **3**B, and **3**C denote the same or similar

elements. The difference between the interlayer connective structures of the printed circuit boards **700**A, **700**B, and **700**C of the comparative examples (including the signal plating through hole structure surrounded by 4 ground plating through hole structures (the second interlayer connective structure **230**) shown in FIG. **4**; the signal plating through hole structure surrounded by 6 ground plating through hole structures (the second interlayer connective structures **230**) shown in FIG. **5**; and the coaxial perforated structure formed by the signal plating through hole structure surrounded by the ground plating through hole structure (the second interlayer connective structure **330**) shown in FIG. **6**) and the interlayer connective structures of the circuit board devices **500**A, **500**B, and **500**C in accordance with some embodiments of the disclosure is at least opposite end surfaces of the interlayer connective structures **710** (also called signal plating through hole structures **710**) of the printed circuit boards **700**A, **700**B, and **700**C of the comparative examples for transmitting signals are directly covered by and directly connected to pads **750**. In addition, the top surfaces of the pads **750** forms portions of the upper surface and portions of the lower surface of the printed circuit boards **700**A, **700**B, and **700**C of the comparative examples. On the other hand, the first interlayer connective structure **210** for transmitting signals (also serve as the signal plating through hole structure **210**) of the circuit board devices **500**A, **500**B, and **500**C in accordance with one embodiment of the disclosure has a pad-free design, the opposite first end surfaces **212**E**1** and **212**E**2** of the first interlayer connective structure **210** are exposed from the substrate **200** and form portions of the upper surface and portions of the lower surface of the circuit board devices **500**A, **500**B, and **500**C. A diameter R**13** of the pads **750** of the printed circuit boards **700**A, **700**B, and **700**C of the comparative examples is in a range of 300 μm to 400 μm, for example, about 350 μm.

[0064] FIG. **7** is a diagram showing the comparison of the characteristic simulation curves of terminal time-domain reflectometry (TDR) characteristic impedance (ohm) versus signal transmission time (ps) between the interlayer connective structures of the circuit board devices **500**A, **500**B and **500**C in accordance with some embodiments of the disclosure and the interlayer connective structures of printed circuit boards **700**A, **700**B, **700**C of the comparative examples.

[0065] In FIG. **7**, the curve **701** shows the characteristic curve of the terminal TDR characteristic impedance versus the signal transmission time of the interlayer connective structure **710** connected to the pads **750** in the printed circuit boards **700**A and **700**B of the comparative examples (that is, the interlayer connective structure **710** surrounded by 4 or 6 ground plating through hole structures and connected to the pads **750** has the same terminal TDR characteristic impedance performance). The curve **702** shows the characteristic curve of the terminal TDR characteristic impedance versus the signal transmission time of the pad-free interlayer connective structure (the first interlayer connective structure) **210** for transmitting signals in the plating through hole structure of the circuit board devices **500**A and **500**B in accordance with some embodiments of the disclosure (that is, the interlayer connective structure (the first interlayer connective structure) **210** surrounded by 4 or 6 ground plating through hole structures has the same characteristic impedance performance). The curve **703** shows the characteristic curve of the terminal TDR characteristic impedance versus the signal transmission time of the interlayer connective structure **710** connected to the pads **750** in the coaxial via structure of the printed circuit board **700**C of the comparative example. The curve **704** shows the characteristic curve of the terminal TDR characteristic impedance versus the signal transmission time of the pad-free interlayer connective structure (the first interlayer connective structure) **210** for transmitting signals in the coaxial via structure of the circuit board device **500**C in accordance with one embodiment of the disclosure.

[0066] It can be seen from the curves **701** and **702** that in the signal plating through hole structure surrounded by the ground plating through hole structures, the conductive path that transmits signals and defined by the signal plating through hole structure having a pad-free design (the curve **702**) may have fewer interfaces, thereby having better impedance matching (impedance continuity). It can be seen from the curves **703** and **704** that in the coaxial via structure, the conductive path that transmits signals and defined by the signal plating through hole structure having a pad-free design (the curve **704**) may have fewer interfaces, thereby having better impedance matching (impedance continuity). It can be seen from the curves **701**, **702**, **703**, and **704** that compared to the signal plating through hole structure surrounded by the ground plating through hole structures (the curves **701**, **702**), the impedance of the signal plating through hole structure of the coaxial via structure (the curves **703**, **704**) is closer to the target impedance (usually of about 50 ohms). Moreover, the signal plating through hole structure in the coaxial via structure having a pad-free design (the curve **704**) has the best impedance matching (impedance continuity), and its characteristic impedance is closest to the target impedance of about 50 ohms.

[0067] FIG. **8** is a diagram showing the comparison of the characteristic simulation curves of reflection coefficient (return loss, Su (dB)) versus signal frequency (GHz) between the interlayer connective structures of the circuit board devices **500**A, **500**B and **500**C in accordance with some embodiments of the disclosure and the interlayer connective structures of printed circuit boards **700**A, **700**B, **700**C of the comparative examples, wherein the signal frequency is in a range of about 1 GHz to 300 GHz.

[0068] In FIG. **8**, the curve **801** shows the characteristic curve of the reflection coefficient (S.sub.11 (dB)) versus the signal frequency (GHz) of the interlayer connective structure **710** connected to the pads **750** in the printed circuit boards **700**A and **700**B of the comparative example (that is, the interlayer connective structure **710** surrounded by 4 or 6 ground plating through hole structures and connected to the pads **750** has the same reflection coefficient performance). The curve **802** shows the characteristic curve of the reflection coefficient versus the signal frequency (GHz) of the pad-free interlayer connective structure (the first interlayer connective structure) **210** for transmitting signals in the plating through hole structure of the circuit board devices **500**A and **500**B in accordance with some embodiments of the disclosure (that is, the interlayer connective structure (the first interlayer connective structure) **210** surrounded by 4 or 6 ground plating through hole structures has the same reflection coefficient performance). The curve **803** shows the characteristic curve of the reflection coefficient versus the signal frequency of the interlayer connective structure **710** connected to the pads **750** in the coaxial via structure of the printed circuit board **700**C of the comparative example. The curve **804** shows the characteristic curve of the reflection coefficient versus the signal frequency of the pad-free interlayer connective structure (the first interlayer connective structure) **210** for transmitting signals in the coaxial via structure of the circuit board device **500**C in accordance with one embodiment of the disclosure.

[0069] It can be seen from the curves **801** and **802** that in the signal plating through hole structure surrounded by the ground plating through hole structures, the signal plating through hole structure having a pad-free design (the curve **702**) has a lower reflection coefficient. It can be seen from the curves **803** and **804** that in the coaxial via structures, the signal plating through hole structure having a pad-free design (the curve **704**) has a lower reflection coefficient. It can be seen from the curves **801**, **802**, **803**, and **804** that the pad-free signal plating through hole structure in the coaxial via structure (the curve **804**) has a reflection coefficient of less than-15 dB at the signal frequency of about 1 GHz to 300 GHz. The reflection coefficient of the pad-free signal plating through hole structure in the coaxial via structure (the curve **804**) at high frequency (about 300 GHz) is about 10 dB lower than the transition region of the printed circuit board in the comparative example (the curve **801**), which can improve the signal reflection loss at high frequencies (about 300 GHz).

[0070] FIG. **9** is a diagram showing the comparison of the characteristic simulation curves of insertion loss (S.sub.21 (dB)) versus signal frequency (GHz) between the interlayer connective structures of the circuit board devices **500**A, **500**B and **500**C in accordance with some embodiments of the disclosure and the interlayer connective structures of printed circuit boards **700**A, **700**B, **700**C of the comparative examples, wherein the signal frequency is in a range of 1 GHz to 300 GHz.

[0071] In FIG. **9**, the curve **901** shows the characteristic curve of the insertion loss versus the signal frequency (GHz) of the interlayer connective structure **710** connected to the pads **750** in the printed circuit boards **700**A and **700**B of the comparative example (that is, the interlayer connective structure **710** surrounded by 4 or 6 ground plating through hole structures and connected to the pads **750** has the same insertion loss performance). The curve **902** shows the characteristic curve of the insertion loss versus the signal frequency of the pad-free interlayer connective structure (the first interlayer connective structure) **210** for transmitting signals in the plating through hole structure of the circuit board devices **500**A and **500**B in accordance with some embodiments of the disclosure (that is, the interlayer connective structure (the first interlayer connective structure **210** surrounded by 4 or 6 ground plating through hole structures has the same insertion loss performance). The curve **903** shows the characteristic curve of the insertion loss versus the signal frequency of the interlayer connective structure **710** connected to the pads **750** in the coaxial via structure of the printed circuit board **700**C of the comparative example. The curve **904** shows the characteristic curve of the insertion loss versus the signal frequency of the pad-free interlayer connective structure (the first interlayer connective structure) **210** for transmitting signals in the coaxial via structure of the circuit board device **500**C in accordance with one embodiment of the disclosure.

[0072] It can be seen from the curves **901** and **903** that the insertion loss curves of the signal plating through hole structures connected to the pads **750** in the printed circuit boards **700**A, **700**B, and **700**C of the comparative examples will produce periodic resonance phenomenon when the signal frequency is above 50 GHz. In each of the curves **901** and **903**, the jitter amplitude of the waveform becomes larger as the frequency increases. Compared with the signal plating through hole structure in the coaxial via structure and connected to the pads (e.g., the printed circuit board **700**C of the comparative example), the signal plating through hole structures surrounded by the ground plating through hole structures and connected to the pads (e.g., the printed circuit boards **700**A, **700**B of the comparative examples) has larger jitter amplitude of the waveform of the insertion loss curve. It can be seen from the curves **901**, **902**, **903**, and **904** that in the coaxial via structure and the signal plating through hole structure that is surrounded by the ground plating through hole structures, the pad-free signal plating through hole structure (the curves **902**, **904**) may improve the jitter amplitude of the waveform of the insertion loss curve and have lower insertion loss. In the signal plating through hole structure surrounded by the ground plating through hole structures, the insertion loss curve of the pad-free signal plating through hole structure (the curve **902**) will not produce periodic resonance until the signal frequencies is above 235 GHZ (for example, the curve **902** produces periodic resonance phenomenon at the signal frequencies of about 235, 270, 290 GHz). Moreover, the pad-free signal plating through hole structure in the coaxial via structure (the curve **904**) does not find the periodic resonance phenomenon in the insertion loss curve at high signal frequencies (the signal frequency is up to about 300 GHz). The pad-free signal plating through hole structure in the coaxial via structure (the curve **904**) may maintain low insertion loss performance at the signal frequency of about 1 GHz to 300 GHz.

[0073] Embodiments of the disclosure provide a circuit board device for transmitting signals at a frequency of 50 GHz to 300 GHz. The circuit board device includes an insulation portion, a first interlayer connective structure, a first reference layer, a second reference layer and a second interlayer connective structure. The first conductive structure defining the signal path has a pad-free design. That is to say, the two first end surfaces of the first conductive structure that passes through the insulation portion are not directly covered by or directly connected to the pads. In other words, the opposite upper and lower surfaces of the circuit board device respectively include opposite first end surfaces. Moreover, the through hole of the first via structure is not sealed by the pads. In some embodiments, the first conductive structure includes a first pillar. Furthermore, the first pillar includes a pair of first end portions and a first central section. In some embodiments, the first wall thickness of the first end portion is greater than a second wall thickness of the first central section. The first reference layer and the second reference layer are located on the top and bottom surfaces of the insulation portion. The first reference layer and the second reference layer are separated from the first interlayer connective structure and surround the first interlayer connective structure. The second interlayer connective structure passes through the insulation portion. Furthermore, two opposite second end surfaces of the second interlayer connective structure are respectively connected to the first reference layer and the second reference layer to define a ground path. The ground path surrounds the signal path.

[0074] In some embodiments, the first interlayer connective structure and the second interlayer connective structure of the circuit board device may form a signal plating through hole structure surrounded by at least one ground plating through hole structure. In this embodiment, the range of the number of first interlayer connective structure is equal to 1. The range of the number of second interlayer connective structure may be greater than or equal to 4. In some embodiments, the first interlayer connective structure and the second interlayer connective structure of the circuit board device may form a coaxial via structure. In this embodiment, the range of the number of first interlayer connective structure and the second interlayer connective structure are both equal to 1.

[0075] In some embodiments, the outer diameter and the wall thickness of the first pillar of the first interlayer connective structure, the outer diameter and the wall thickness of the second pillar of the second interlayer connective structure, the inner and outer diameters of the first reference layer and the second reference layer, the diameters of the first through hole accommodating the first interlayer connective structure and the second through hole accommodating the second interlayer connective structure and distance between the first through hole and the second through hole may be selected according to the process ability and electrical analysis results to further improve the impedance matching with the transmission line, the reflection coefficient and the insertion loss.

[0076] Since the circuit board device in accordance with one embodiment of the disclosure includes the signal plating through hole structure having a pad-free design, the conductive path for transmitting signals and defined by the pad-free signal plating through hole structure has fewer interfaces. The conductive path may be formed with low variation of impedance. The impedance matching (impedance continuity) with the transmission line can be improved. The return loss due to the impedance mismatch when transmitting high-frequency signals can be reduced. Moreover, the pad-free signal plating through hole structure in the coaxial via structure has the best impedance matching (impedance continuity), and its characteristic impedance is closest to the target impedance of about 50 ohms. In addition, the pad-free signal plating through hole structure of the circuit board device in accordance with one embodiment of the disclosure has a lower reflection coefficient. Moreover, the pad-free signal plating through hole structure in the coaxial via structure has a reflection coefficient of less than about-15 dB at the signal frequency of about 1 GHz to 300 GHz. The reflection coefficient of the pad-free signal plating through hole structure in the coaxial via structure at high frequency (about 300 GHz) is about 10 dB lower than the transition region of the printed circuit board in the comparative example, which can improve the signal reflection loss at high frequencies (about 300 GHz). In addition, the signal plating through hole structure having a pad-free design in accordance with one embodiment of the disclosure may improve the jitter amplitude of the waveform of the insertion loss curve and have lower insertion loss. In the signal plating through hole structure surrounded by the ground plating through hole structures, the insertion loss curve of the pad-free signal plating through hole structure will not produce periodic resonance until the signal frequencies is above 235 GHz. Moreover, the pad-free signal plating through hole structure in the coaxial via structure does not find the periodic resonance phenomenon in the insertion loss curve at high signal frequencies (the signal frequency is up to about 300 GHz). The pad-free signal plating through hole structure in the coaxial via structure may maintain low insertion loss performance at the signal frequency of about 1 GHz to 300 GHz. The circuit board device in accordance with one embodiment of the disclosure may have a reduced loss during current signal transmission. Therefore, the signal transmission quality of the circuit board can be improved.

[0077] While the invention has been described by way of example and in terms of the preferred embodiments, it should be understood that the invention is not limited to the disclosed embodiments. On the contrary, it is intended to cover various modifications and similar arrangements (as would be apparent to those skilled in the art). Therefore, the scope of the appended claims should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements.

Claims

1. A circuit board device for transmitting signals at a frequency of 50 GHz to 300 GHz, comprising: an insulation portion having a top surface and a bottom surface; a first interlayer connective structure passing through the insulation portion to define a signal path, wherein the first interlayer connective structure comprises: a first pillar passing through the insulation portion and having two opposite first end surfaces, wherein the first pillar comprises: a pair of first end portions having respective first end surfaces; and a first central section connected between the first end portions, wherein a first wall thickness of each of the first end portions is greater than a second wall thickness of the first central section; a first reference layer located on the top surface of the insulation portion, wherein the first reference layer is separated from the first interlayer connective structure and surrounds the first interlayer connective structure; a second reference layer located on the bottom surface of the insulation portion, wherein the second reference layer is separated from the first interlayer connective structure and surrounds the first interlayer connective structure; and at least one second interlayer connective structure passing through the insulation portion, wherein two opposite second end surfaces of the second interlayer connective structure are respectively connected to the first reference layer and the second reference layer to define a ground path that surrounds the signal path, wherein an upper surface and a lower surface of the circuit board device comprise the respective first end surfaces.
2. The circuit board device as claimed in claim 1, wherein in a top view, the first reference layer and the second reference layer are annular and completely overlap each other, and a first circle center of the first reference layer and a second circle center of the second reference layer both overlap a first pillar axis of the first pillar.
3. The circuit board device as claimed in claim 1, wherein the first pillar has a tubular shape and has a first pillar inner diameter, and the first pillar inner diameter is in a range of 50μ m to 150 μm.
4. The circuit board device as claimed in claim 1, wherein the minimum value of the first wall thickness and the minimum value of the second wall thickness are each 25 μm.
5. The circuit board device as claimed in claim 1, wherein the second interlayer connective structure comprises: a second pillar passing through the insulation portion, wherein the second pillar comprises: a pair of second end portions having respective second end surfaces; and a second central section connected between the second end portions, wherein a third wall thickness of each of the second end portions is greater than a fourth wall thickness of the second central section.
6. The circuit board device as claimed in claim 5, wherein the minimum value of the third wall thickness and the minimum value of the fourth wall thickness are both 25 μm.
7. The circuit board device as claimed in claim 5, wherein: each of the first interlayer connective structure and the second interlayer connective structure is a plating through hole structure, the first interlayer connective structure is located in a first through hole passing through the insulation portion, and the first pillar covers an inner wall of the first through hole; and the second interlayer connective structure is located in a second through hole passing through the insulation portion, and the second pillar covers an inner wall of the second through hole.
8. The circuit board device as claimed in claim 7, wherein the first through hole has a first diameter, and the first diameter is in a range of 100 μm to 200 μm.
9. The circuit board device as claimed in claim 7, wherein the second through hole has a second diameter, and the second diameter is in a range of 100 μm to 200 μm.

10. The circuit board device as claimed in claim 7, wherein the second through hole has a second diameter, and the second diameter is in a range of 100 μm to 200 μm.

11. The circuit board device as claimed in claim 7, wherein the first reference layer and the second reference layer respectively have a first reference layer inner diameter and a second reference layer inner diameter, and the first reference layer inner diameter and the second reference layer inner diameter are in a range of 500 μm to 600 μm.

12. The circuit board device as claimed in claim 11, wherein the first reference layer and the second reference layer respectively have a first reference layer outer diameter and a second reference layer outer diameter, and the first reference layer outer diameter and the second reference layer outer diameter are in a range of 1200 μm to 1300 μm.

13. The circuit board device as claimed in claim 7, wherein the first through hole is spaced apart from the second through hole by a first distance, and the first distance is in a range of 250 μm to 350 μm.

14. The circuit board device as claimed in claim 7, wherein in a top view, the second through hole is spaced apart from a first inner edge of the first reference layer by a second distance, and the second distance is in a range of 50 μm to 150 μm.

15. The circuit board device as claimed in claim 7, wherein in a top view, the second through hole is spaced apart from a first outer edge of the first reference layer by a third distance, and the third distance is in a range of 50 μm to 150 μm.

16. The circuit board device as claimed in claim 7, wherein the first reference layer and the second reference layer surround a first number of first interlayer connective structures, and the first reference layer and the second reference layer are connected to a second number of second interlayer connective structures, wherein the first number is equal to 1, and the second number is greater than or equal to 4.

17. The circuit board device as claimed in claim 16, wherein second through-hole axes of the second through holes are located on a circle whose center is located at a first through-hole axis of the first through hole, and a central angle between two radii connecting the second through-hole axes of the two closest second through holes to the first through-hole axis of the first through hole is less than or equal to 90 degrees.

18. The circuit board device as claimed in claim 5, wherein the first interlayer connective structure and the second interlayer connective structure form a coaxial via structure, and the second interlayer connective structure is located in a third through hole passing through the insulation portion, wherein the second pillar covers an inner wall of the third through hole, and the coaxial via structure further comprises: a first dielectric material filling the third through hole, wherein the second pillar surrounds the first dielectric material, and the first interlayer connective structure is located in a fourth through hole passing through the first dielectric material.

19. The circuit board device as claimed in claim 18, wherein the first reference layer and the second reference layer respectively partially cover a top surface and a bottom surface of the first dielectric material.

20. The circuit board device as claimed in claim 18, wherein the first reference layer and the second reference layer surround a first number of first interlayer connective structures, and the first reference layer and the second reference layer are connected to a second number of second interlayer connective structures, wherein the first number and the second number are both equal to 1.

21. The circuit board device as claimed in claim 18, wherein the third through hole and the fourth through hole are substantially coaxial.

22. The circuit board device as claimed in claim 18, wherein the third through hole has a third diameter, and the third diameter is in a range of 650 μm to 750 μm.

23. The circuit board device as claimed in claim 18, wherein the fourth through hole has a fourth diameter, and the third diameter is in a range of 650 μm to 750 μm.

24. The circuit board device as claimed in claim 18, wherein the second pillar has a tubular shape and has a second pillar inner diameter, and the second pillar inner diameter is in a range of 600 μm to 700 μm.

25. The circuit board device as claimed in claim 18, wherein the second pillar has a tubular shape and has a second pillar inner diameter, and the second pillar inner diameter is in a range of 600 μm to 700 μm.

26. The circuit board device as claimed in claim 25, wherein each of the first reference layer and the second reference layer has a reference layer outer diameter, and the reference layer outer diameter is in a range of 650 μm to 750 μm.

27. The circuit board device as claimed in claim 18, wherein the third through hole is spaced apart from the fourth through hole by a fourth distance, and the fourth distance is in a range of 225 μm to 325 μm.

28. The circuit board device as claimed in claim 18, wherein an impedance of the coaxial via structure is in a range of 45 ohms to 55 ohms.

29. The circuit board device as claimed in claim 1, wherein a thickness of the insulation portion is in a range of 2.09 mm to 2.29 mm.